

EQUIPMENT: UNIT 7 SUPERCRITICAL STEAM TURBINE

REPORT ID: QA-NDT-2026-08892

LOCATION: BAY-4B HIGH PRESSURE GENERATION CASING

STANDARD: SOP-TURB-IND-2026-V4

INSPECTION DATE: 2026-09-18 | CLEARANCE: RESTRICTED

SENSOR CALIBRATION: NPL-TRACEABLE 2026

SECTION 1: SENSOR TELEMETRY & PHYSICAL METROLOGY READINGS

PARAMETER NAME	PROBE ID	MEASURED VALUE	SOP THRESHOLD	METROLOGY STATUS
Radial Vibration (Drive End RMS)	VIB-DE-701	4.85 mm/s	<= 3.50 mm/s	NON_COMPLIANT (EXCEEDED)
Radial Vibration (Non-Drive End)	VIB-NDE-702	2.80 mm/s	<= 3.50 mm/s	COMPLIANT
Journal Bearing #2 Temperature	TH-JB-201	98.4 deg C	<= 85.0 deg C	NON_COMPLIANT (CRITICAL)
Bearing Lube Oil Return Pressure	PRS-LO-401	0.78 bar	>= 1.20 bar	NON_COMPLIANT (DEFICIT)
Rotor Axial Displacement	DISP-AX-101	0.42 mm	<= 0.60 mm	COMPLIANT

SECTION 2: OPTICAL BORESCOPE & DEFECT SURFACE ANALYSIS

1. High-pressure turbine stage 3 rotor blades exhibit severe cavitation pitting and micro-fractures.

2. Cavitation depth measured at 1.42 mm across leading edge root fillet (Threshold: <= 0.50 mm per DEF-STD-05-21).

3. White-metal Babbitt lining on Journal Bearing #2 shows thermal wipe streaks and metallic debris scoring.

4. Risk assessment: High probability of catastrophic rotor-stator rub if operated above 1500 RPM.

INSPECTOR SIGN-OFF & FIELD CERTIFICATION

Officer Name: Col. V. Sharma / Senior Inspecting Officer

Signature: [Col. V. Sharma - Sovereign Seal #88192]

Action Required: Submit Official Approval Note Sheet (.docx)

QA INSPECTION VERDICT: REJECTED

STATUS: IMMEDIATE EMERGENCY OVERHAUL MANDATED

REF: DEFENCE PROCUREMENT NOTE TURB-07/2026

VERIFIED BY: CHIEF QUALITY AUDITOR (SEALED)